

User's most recent purchases (chronological)

1. Neutrogena Ultra Sheer
Dry-Touch Sunscreen SPF 45

2. CeraVe Foaming
Facial Cleanser

3. The Ordinary Niacinamide
10% + Zinc 1% Serum

Query text Q
(titles 1+2+3 concatenated)

Same query Q , two possible
retrieval outcomes at rank 1 →

(a) What a well-trained retriever should return

#1 CeraVe Moisturizing Cream

✓ not in history
✓ same skincare routine
(moisturizing step)

#2 Aveeno Daily Moisturizing Lotion

#3 La Roche-Posay Toleriane Moisturizer

(b) What actually happens in practice — history contamination

#1 CeraVe Foaming Facial Cleanser

✗ already item 2
in the query itself
(re-recommended)

#2 CeraVe Moisturizing Cream (correct answer, pushed down)

#3 Neutrogena Hydro Boost Gel Cream